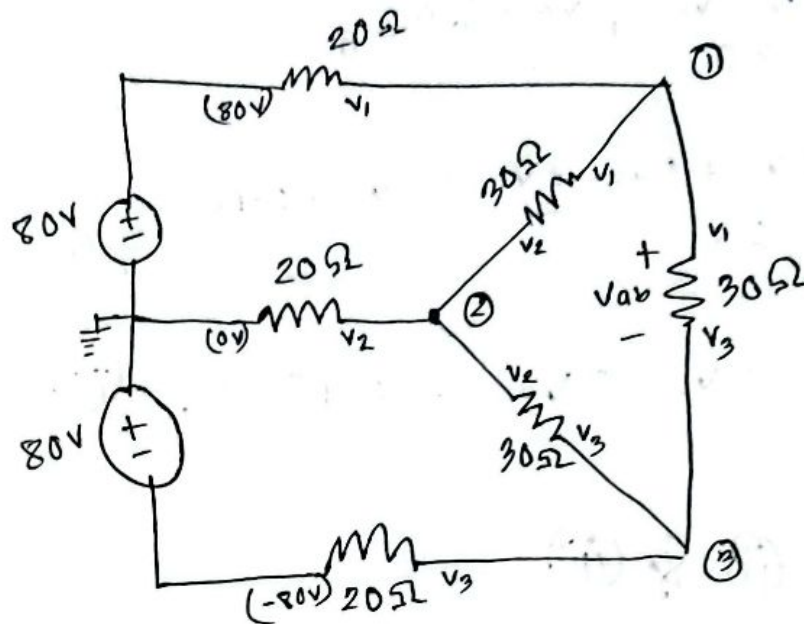


1.



At Node-1:

$$\frac{V_1 - 80}{20} + \frac{V_1 - V_2}{30} + \frac{V_1 - V_3}{30} = 0$$

$$\Rightarrow \frac{3V_1 - 240 + 2V_1 - 2V_2 + 2V_1 - 2V_3}{60} = 0$$

$$\Rightarrow 7V_1 - 2V_2 - 2V_3 = 240 \quad \text{--- (i)}$$

Node-2:

$$\frac{V_2 - 0}{20} + \frac{V_2 - V_1}{30} + \frac{V_2 - V_3}{30} = 0$$

$$\Rightarrow \frac{3V_2 + 2V_2 - 2V_1 + 2V_2 - 2V_3}{60} = 0$$

$$\Rightarrow -2V_1 + 7V_2 - 2V_3 = 0 \quad \text{--- (ii)}$$

Node-3:

$$\frac{V_3 - V_1}{30} + \frac{V_3 - V_2}{30} + \frac{V_3 + 80}{20} = 0$$

$$\Rightarrow \frac{2V_3 - 2V_1 + 2V_3 - 2V_2 + 3V_3 + 240}{60} = 0$$

$$\Rightarrow -2V_1 - 2V_2 + 7V_3 = -240 \quad \text{--- (ii)}$$

Solving (i), (ii) & (iii)

$$V_1 = 26.66 \text{ V}$$

$$V_2 = 0 \text{ V}$$

$$V_3 = -26.66 \text{ V}$$

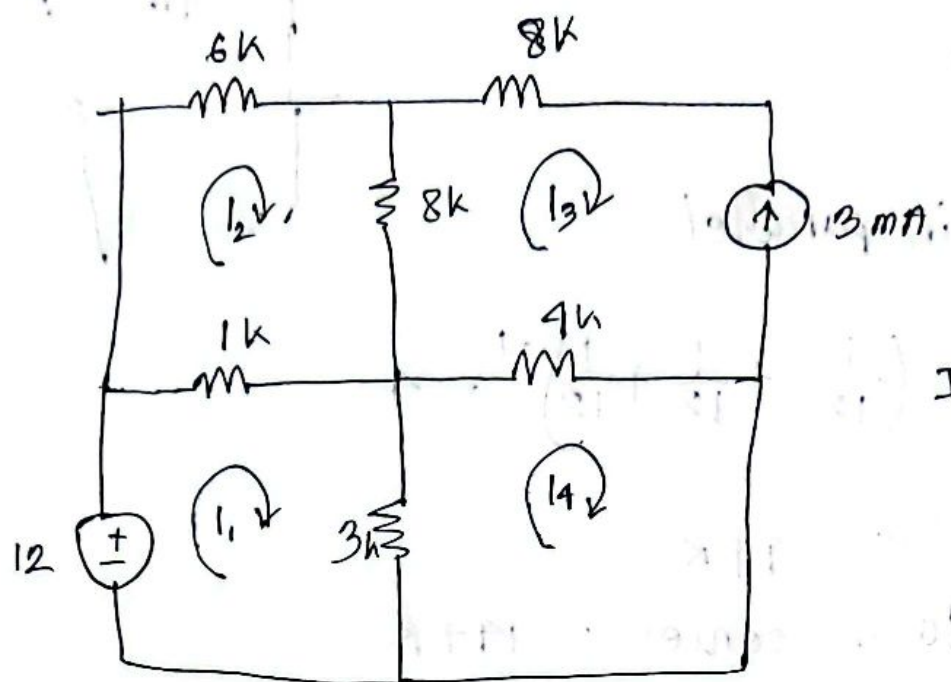
$$V_{ab} = V_1 - V_3 = 53.32 \text{ V}$$

$$I_o = \frac{V_{ab}}{R} = 1.77 \text{ A}$$

$$V_x = V_2 = 0 \text{ V}$$

$$I_x = \frac{V}{R} = \frac{V_1 - V_2}{R} = \frac{26.66}{30} = \cancel{1.77 \text{ A}} 0.88 \text{ A}$$

2



$$I_3 = -3 \text{ mA}$$

loop-1:

$$-12 + 1(I_1 - I_2) + 3(I_1 - I_4) = 0$$

$$\Rightarrow -12 + I_1 - I_2 + 3I_1 - 3I_4 = 0$$

$$\Rightarrow 4I_1 - I_2 - 3I_4 = 12 \quad \text{--- (i)}$$

loop-2:

$$6I_2 + 8(I_2 - I_3) + 1(I_2 - I_1) = 0$$

$$\Rightarrow 6I_2 + 8I_2 + 24 + I_2 - I_1 = 0$$

$$\Rightarrow -I_1 + 15I_2 = -24 \quad \text{--- (ii)}$$

loop-4:

$$3(I_4 - I_1) + 4(I_4 - I_3) = 0$$

$$\Rightarrow 3I_4 - 3I_1 + 4I_4 + 12 = 0$$

$$\Rightarrow -3I_1 + 7I_4 = -12$$

$$I_3 = -3 \text{ mA}$$

$$I_1 = 1.08$$

$$I_2 = -1.46$$

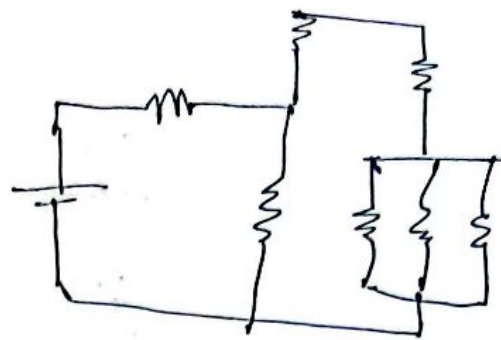
$$I_4 = -0.86$$

वाकिछे साविना

③ a

$$R_{eq} = 50$$

3, 12 Ω in parallel



$$\therefore R_p = \left(\frac{1}{12} + \frac{1}{12} + \frac{1}{12} \right)^{-1} = 4$$

R is series = 4 + R

10 Ω is also in series = 14 + R

60 Ω in parallel with them

$$\therefore R_p = \left(\frac{1}{14+R} + \frac{1}{60} \right)^{-1}$$

and last 10 Ω in series

$$\therefore R_{eq} = \left(\frac{1}{14+R} + \frac{1}{60} \right)^{-1} + 10 = 50$$

$$\Rightarrow \left(\frac{60 + 14 + R}{60(14+R)} \right)^{-1} = 40$$

$$\Rightarrow \frac{60(14+R)}{74+R} = 40$$

$$\cancel{74+R} = \cancel{2400R} + \cancel{33600}$$

$$\cancel{2326R} =$$

$$\Rightarrow 2260 + 40R = 840 + 60R$$

$$\Rightarrow 20R = 2120$$

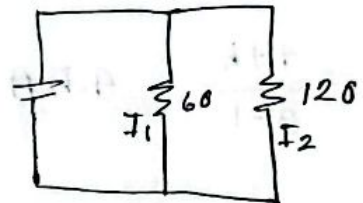
$$\therefore R = 106 \Omega$$

b.

main current

$$I = \frac{20}{50} = 0.4 \text{ A}$$

$$\therefore I_2 = \frac{60}{180} = 0.33 \text{ A}$$



$\therefore$ Voltage V across 12Ω is $= 0.33 \times 4 = 1.33 \text{ V}$
 current through each 12Ω resistor is $= \frac{4}{12} \times 0.33$
 $= 0.11$

summer-23

$$\textcircled{1} \quad i = \frac{dq}{dt}$$

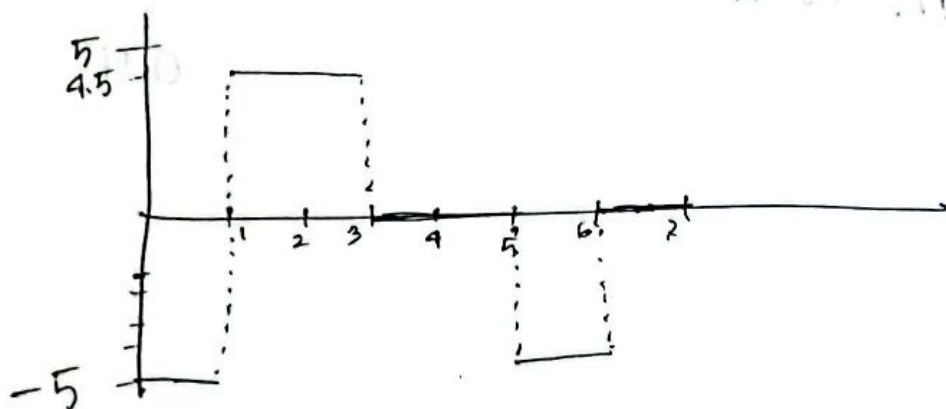
$$i_1 = \frac{-5-0}{1} = -5A \quad ; \quad 0 < t < 1$$

$$i_2 = \frac{4+5}{3-1} = 4.5A \quad ; \quad 1 < t < 3$$

$$i_3 = 0A \quad ; \quad 3 < t < 5$$

$$i_4 = \frac{-5-0}{1} = -5A \quad ; \quad 5 < t < 6$$

$$i_5 = 0A \quad ; \quad 6 < t < 7$$



current at 2.5 is 4.5A

" " 5.5 is -5A

(i) $P = I^2 R$

$\therefore P_1 = I_1^2 R = 500 \text{ W} ; 0 < t < 1$

$P_2 = I_2^2 R = 405 \text{ W} ; 1 < t < 3$

$P_3 = I_3^2 R = 0 \text{ W} ; 3 < t < 5$

$P_4 = I_4^2 R = 500 \text{ W} ; 5 < t < 6$

$P_5 = I_5^2 R = 0 \text{ W} ; 6 < t < 7$

at $0 < t < 1$ & $5 < t < 6$ maximum power delivered to the element. maximum power is 500 W.

2.
cd || cd

$\therefore R_{cd} = \left(\frac{1}{15} + \frac{1}{20} \right)^{-1} = 4 \Omega$

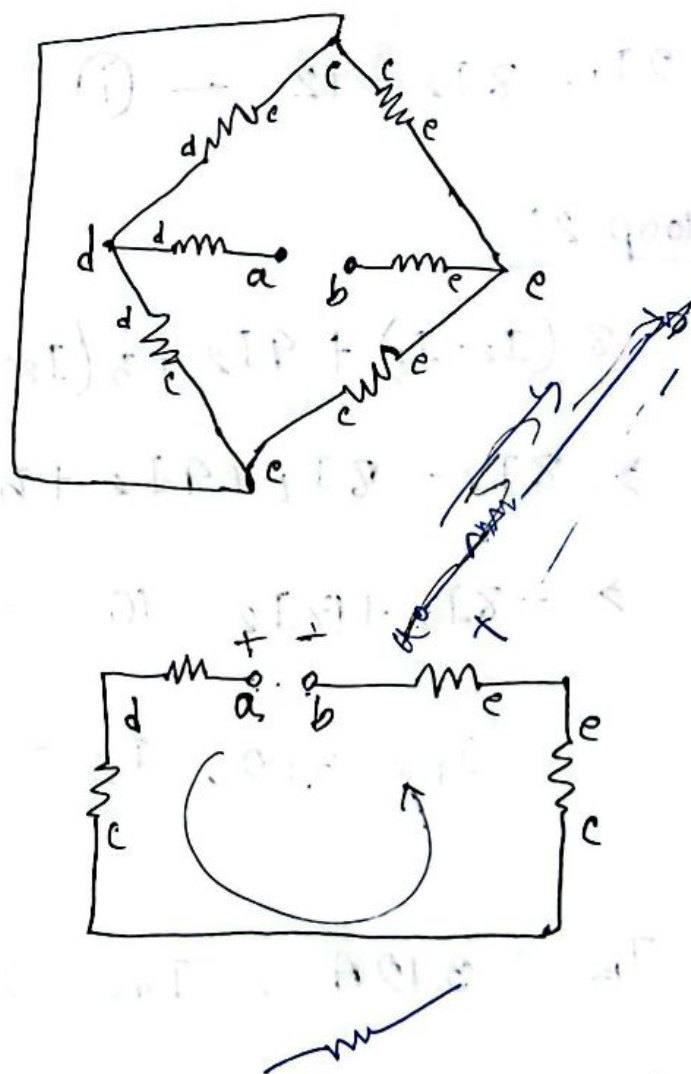
ce || ce

$\therefore R_{ce} = \left(\frac{1}{6} + \frac{1}{3} \right)^{-1} = 2 \Omega$

All in series

$\therefore R_{ab} = 10 + 4 + 2 + 8$
 $= 24 \Omega$

(ii)
 $R_{ec} =$



(ii) Total current $I = \frac{30}{24} = 1.25 A$

$\therefore I_{ef} = \frac{3}{3+6} \times 1.25 = 0.41 A$

$I_{eb} = 1.25 A$

3

loop-1:

$-12 + I_1 + 8(I_1 - I_2) = 0$

$9I_1 - 8I_2 = 12 \quad \text{--- (i)}$

loop-2:

$8(I_2 - I_1) + 4I_2 + 5(I_2 - I_3) = 0$

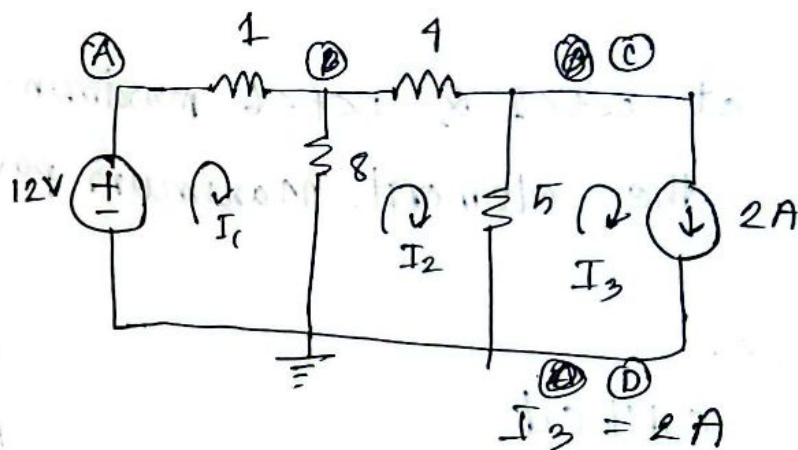
$\Rightarrow 8I_2 - 8I_1 + 4I_2 + 5I_2 - 5I_3 = 0$

$\Rightarrow -8I_1 + 17I_2 = 10 \quad \text{--- (ii)}$

$I_1 = 3.19, I_2 = 2.08, I_3 = 2$

$I_{1\Omega} = 3.19 A, I_{4\Omega} = 2.08 A, I_{2A} = 2 A$

$I_{12V} = 3.19 A, I_{8\Omega} = I_1 - I_2 = 1.11, I_{5\Omega} = I_2 - 2 = 0.08 A$

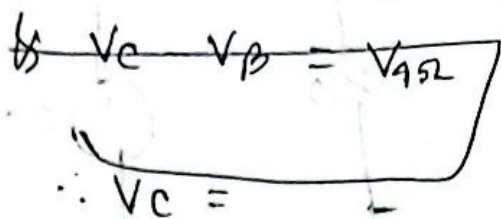


Node - D voltage is 0V

Node - A " " " 12V

$$V_A - V_B = V_{1\Omega}$$

$$\therefore V_B = \cancel{3.12} V_A - V_{1\Omega} = 12 - 3.12 = 8.8V$$



$$V_B - V_C = V_{1\Omega}$$

$$\therefore V_C = V_B - \cancel{4} V_{1\Omega}$$

$$= 8.8 - (4 \times 2.08)$$

$$= 0.48V$$

④

loop-1:

$$30I_1 + 10(I_1 - I_2) + 120 = 0$$

$$\Rightarrow 30I_1 + 10I_1 - 10I_2 + 120 = 0$$

$$40I_1 - 10I_2 + 120 = 0 \quad \text{--- (i)}$$

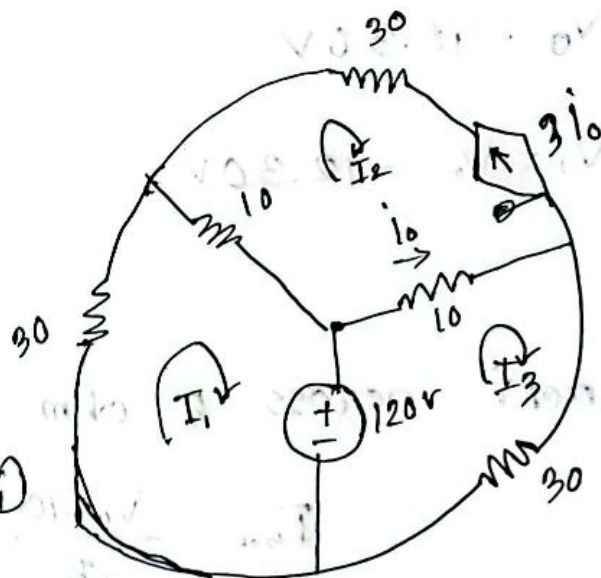
loop-2: $-2I_2 + 3I_3 = 0 \quad \text{--- (ii)}$

loop-3:

$$-120 + 10(I_3 - I_2) + 30I_3 = 0$$

$$10I_3 - 10I_2 + 30I_3 = 120$$

$$40I_3 - 10I_2 = 120 \quad \text{--- (iii)}$$



$$I_2 = -3i_o$$

$$I_2 = -3i_o$$

$$i_o = I_3 - I_2$$

$$\therefore I_2 = -3(I_3 - I_2)$$

$$I_2 = -3I_3 + 3I_2$$

$$I_1 = -1.2$$

$$I_2 = 7.2$$

$$I_3 = 4.8$$

$$\therefore I_0 = 4.8 - 7.2 \\ = -2.4 \text{ A}$$

(5)

$$\frac{V_0 - 10}{10} + \frac{V_0 - 10}{5} + \frac{V_0}{10} + \frac{V_0 - 20}{4} = 0$$

$$\Rightarrow \frac{2V_0 - 20 + 4V_0 - 40 + 2V_0 + 5V_0 - 100}{20} = 0$$

$$\Rightarrow 13V_0 = 160$$

$$\therefore V_0 = 12.30 \text{ V}$$

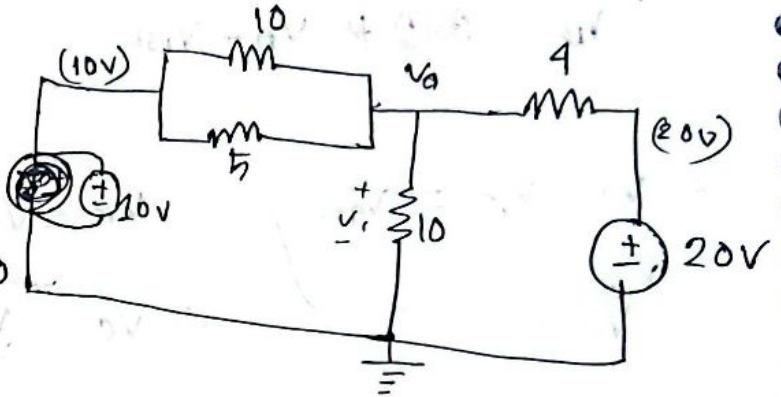
$$\therefore V_1 = V_0 = 12.30 \text{ V}$$

Current across 5 ohm resistor is

$$I_{5\Omega} = \frac{V_0 + 10}{5} = \frac{12.30 + 10}{5} = \frac{22.30}{5}$$

$$= 0.46 \text{ A}$$

(ii) 20V, because it is higher and V_0 voltage is 12.3V and it flows through high to low voltage. so it flows through 12.3V to 10V



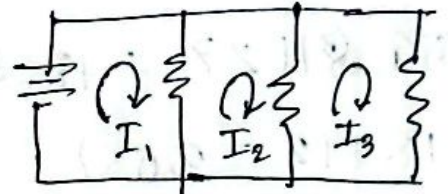
summer 24

① any assignment 4 number same.

② using KVL

$$\begin{array}{l|l} -10 + V_1 - 6 + 2 + 3 = 0 & -10 + V_2 + 3 = 0 \\ V_1 = 11 \text{ V} & \therefore V_2 = 7 \text{ V} \end{array}$$

ii) Normally $I_s = 8 + 6 = 14 \text{ mA}$
 $I_2 = 4 \text{ mA}$



KVL at loop 2 & 3

$$I_2 = 6 \text{ mA}, I_3 = 2 \text{ mA}$$

$$\begin{cases} R_1(I_1 - I_2) = 8 \\ R_1 I_1 - R_1 I_2 = 8 \end{cases}$$

loop-1

$$I_1 - I_2 = 8$$

$$I_1 = 8 + I_2 = 8 + 6 = 14 \text{ mA}$$

$$I_2 = I_1 - I_3$$

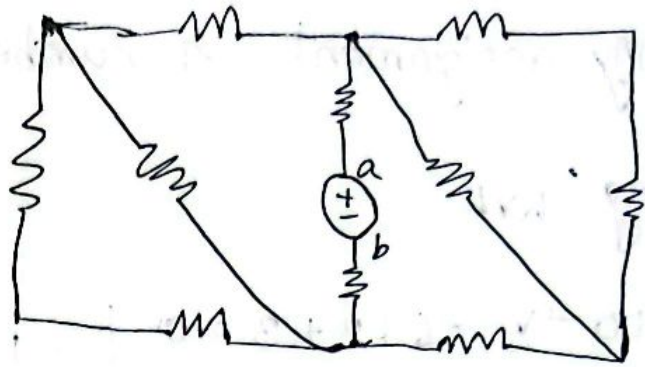
$$= 14 - 10 = 4 \text{ mA}$$

③ (i) left loop

$$50 + 30 = 80 \Omega$$

$$80 \parallel 20 = 16$$

$$16 + 14 = 30 \Omega$$



right loop

$$30 + 24 = 54 \Omega$$

$$54 \parallel 27 = 18 \Omega$$

$$18 + 12 = 30 \Omega$$

$$30 \parallel 30 = 15$$

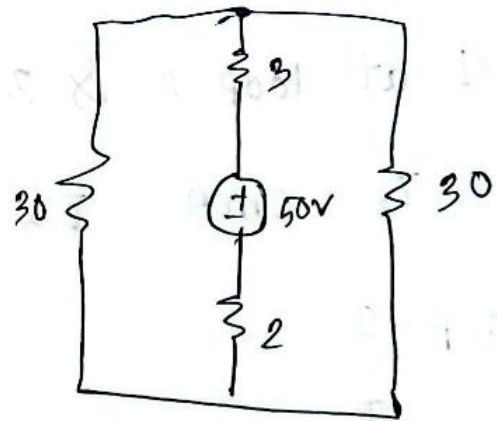
$$\therefore R_{eq} = 15 + 3 + 2 = 20 \Omega$$

(ii) Main current $I = \frac{50}{20} = 2.5 \text{ A}$

Both side goes $\frac{2.5}{2} = 1.25 \text{ A}$

$$\therefore I_2 = \frac{54}{54 + 27} \times 1.25 = 0.83 \text{ A}$$

$$I_1 = \frac{20}{20 + 80} \times 1.25 = 0.25 \text{ A}$$

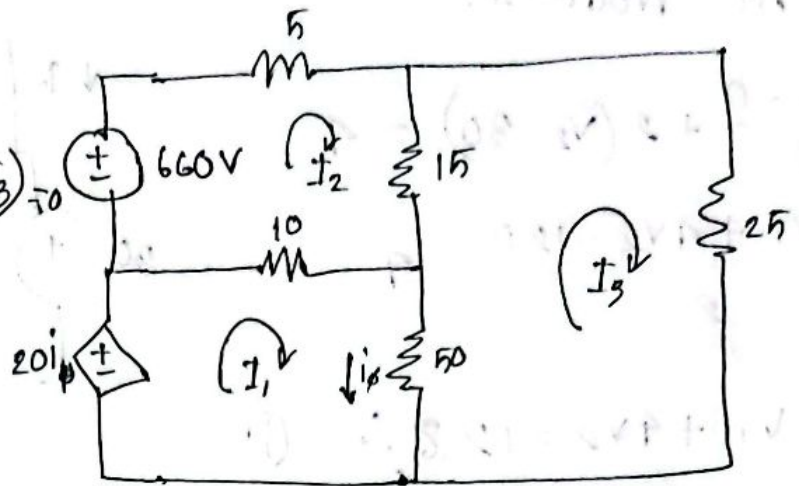


$$\text{Current at } V_1 = \frac{27}{54 + 27} \times 1.25 = 0.41 \text{ A}$$

$$\therefore \text{voltage at } V_1 = 0.41 \times 30 = 12.5 \text{ V}$$

④ loop-1:

$$-20i_\phi + 10(I_1 - I_2) + 50(I_1 - I_3) = 0$$



$$\Rightarrow -20i_\phi + 10I_1 - 10I_2 + 50I_1 - 50I_3 = 0$$

$$\Rightarrow -20I_1 + 20I_3 + 10I_1 - 10I_2 + 50I_1 - 50I_3 = 0$$

$$\Rightarrow \cancel{50I_1} - 10I_2 + 40I_1 - 10I_2 - 30I_3 = 0 \quad \text{--- (i)}$$

$$I_0 = I_1 - I_3$$

loop-2:

$$-660 + 5I_2 + 15(I_2 - I_3) + 10(I_2 - I_1) = 0$$

$$\Rightarrow -660 + 5I_2 + 15I_2 - 15I_3 + 10I_2 - 10I_1 = 0$$

$$\Rightarrow -10I_1 + 30I_2 - 15I_3 = 660 \quad \text{--- (ii)}$$

loop-3:

$$15(I_3 - I_2) + 25I_3 + 50(I_3 - I_1) = 0$$

$$\Rightarrow 15I_3 - 15I_2 + 25I_3 + 50I_3 - 50I_1 = 0$$

$$\Rightarrow -50I_1 - 15I_2 + 90I_3 = 0 \quad \text{--- (iii)}$$

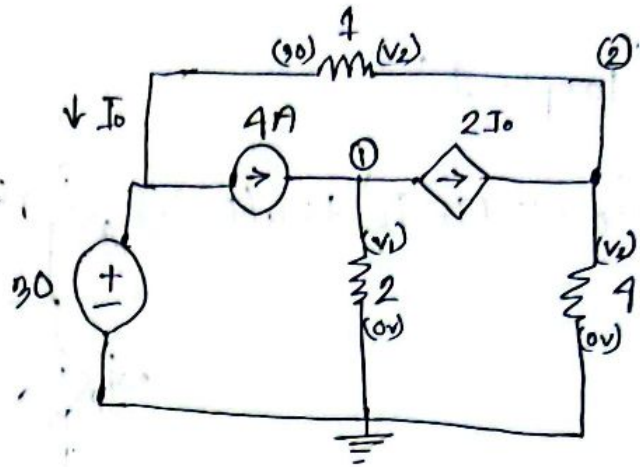
$$\therefore I_1 = 27, I_2 = 42, I_3 = 22 \quad \therefore I_0 = 27 - 22 = 5A$$

(ii) At Node-1:

$$\frac{V_1 - 0}{2} + 2(V_2 - 30) = 4$$

$$\Rightarrow \frac{V_1 + 4V_2 - 120}{2} = 4$$

$$\Rightarrow V_1 + 4V_2 = 128 \quad \text{--- (i)}$$



$$I_0 = \frac{V_2 - 30}{1}$$

Node-2:

$$\frac{V_2 - 30}{1} + \frac{V_2 - 0}{4} = 2(V_2 - 30)$$

$$V_2 - 30 + \frac{V_2}{4} - 2V_2 = -60 + 30$$

$$\Rightarrow \frac{4V_2 + V_2 - 8V_2}{4} = -30$$

$$\Rightarrow -3V_2 = -120$$

$$\therefore V_2 = 40V$$

$$\therefore V_1 = 128 - 4 \times 40 = -32V$$

$$\therefore I_0 = \frac{40 - 30}{1} = 10A$$

spring 24

1. my assignment 5 numbers

$$\textcircled{2} = -20 + 2I_1 - 30 + 2(I_1 - I_2) = 0$$

①

$$\Rightarrow 2I_1 + 2I_1 - 2I_2 = 50$$

$$\Rightarrow 4I_1 - 2I_2 = 50 \quad \text{--- } \textcircled{i}$$

loop LA

loop-2: $2(I_2 - I_1) + 30 + 1I_2 + 4V_x + 1I_2 = 0$

$$\Rightarrow 2I_2 - 2I_1 + 30 + 2I_2 + 4(2I_1) = 0$$

$$V_x = 2I_1$$

$$\Rightarrow 6I_1 + 4I_2 = -30 \quad \text{--- } \textcircled{ii}$$

① solving ① & ②

$$I_1 = 5A \quad I_2 = -15A$$

$$I_2 = -3I_1$$

$$\Rightarrow I_1 = \frac{-15}{-3} = 5A$$

$$V_x = 2I_1$$

$$\therefore V_x = 10V$$

③

$$BC \parallel BC = \frac{R}{2}$$

$$CD_1 = \frac{R}{2} + \frac{R}{2} = R$$

$$CD_2 = \frac{R}{2} + \frac{R}{2} = R$$

$$\therefore CD \parallel CD \parallel CD = \frac{R}{3}$$

$$BD = \frac{R}{3} + \frac{R}{2} = \frac{5R}{6}$$

$$BD \parallel BD = R + \frac{5R}{6}$$

$$\therefore AB = R + R + \frac{5R}{6} = 10$$

$$\Rightarrow \frac{10R + 6}{5R} = 10$$

$$\Rightarrow 10R - 50R + 6 = 0$$

$$\Rightarrow 7R = 60$$

$$\Rightarrow R = \frac{60}{7}$$

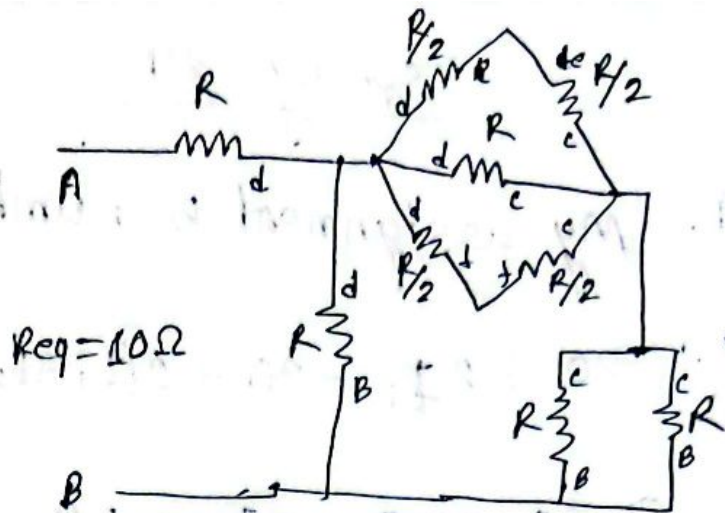
$$BD \parallel BD = \left(\frac{1}{R} + \frac{1}{5R/6} \right)^{-1} = \left(\frac{1}{R} + \frac{6}{5R} \right)^{-1}$$

$$= \left(\frac{5 + 6}{5R} \right)^{-1} = \frac{5R}{11}$$

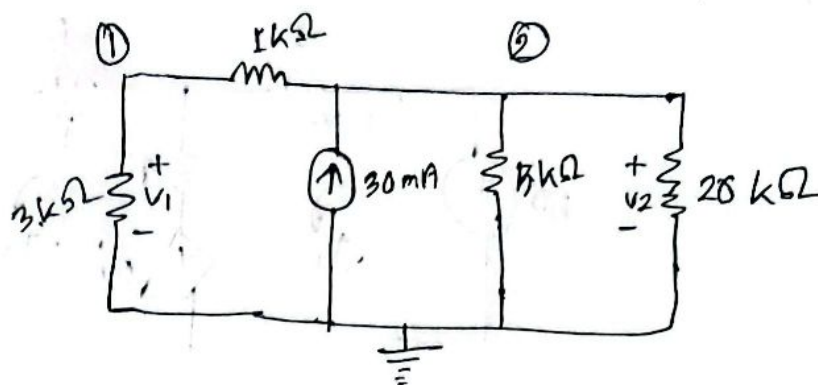
$$\therefore AB = R + \frac{5R}{11} = \frac{11R + 5R}{11} = \frac{16R}{11}$$

$$\therefore \frac{16R}{11} = 10 \Rightarrow 16R = 110$$

$$\therefore R = 6.875 \Omega$$



⑪ Nodal Analysis



Node-1

$$\frac{V_1 - 0}{3} + \frac{V_1 - V_2}{1} = 0$$

$$\frac{V_1 + 3V_1 - 3V_2}{3} = 0$$

$$\therefore 4V_1 - 3V_2 = 0 \quad \text{--- (1)}$$

Node-2

$$\frac{V_2 - V_1}{1} + \frac{V_2 - 0}{5} + \frac{V_2 - 0}{20} = 30$$

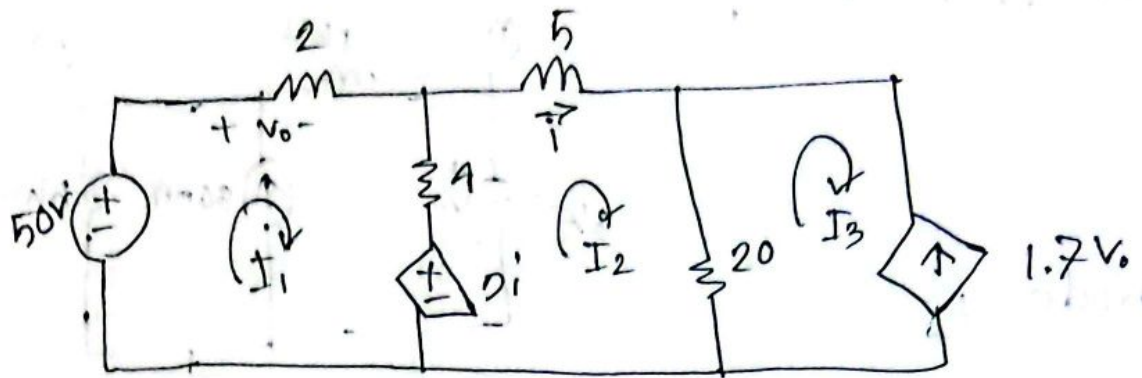
$$\Rightarrow \frac{20V_2 - 20V_1 + 4V_2 + V_2}{20} = 30$$

$$\Rightarrow -20V_1 + 25V_2 = 600 \quad \text{--- (2)}$$

$$V_1 = 45V, V_2 = 60V$$

$$\therefore V_1 = 45V, V_2 = 60V$$

④ ①



loop-1:

$$-50 + 2I_1 + 4(I_1 - I_2) + 2i = 0$$

$$\Rightarrow -50 + 2I_1 + 4I_1 - 4I_2 + 2I_2 = 0$$

$$\Rightarrow 6I_1 + 5I_2 = 50 \quad \text{--- (i)}$$

$$I_3 = -1.7V_o$$

$$V_o = I_1 \times 2 = 2I_1$$

$$\therefore I_3 = -3.4I_1$$

$$i = I_2$$

loop-2:

$$-2I_2 + 4(I_2 - I_1) + 5I_2 + 20(I_2 - I_3) = 0$$

$$\Rightarrow -2I_2 + 4I_2 - 4I_1 + 5I_2 + 20I_2 - 20I_3 = 0$$

$$\Rightarrow -4I_1 + 20I_2 - 20I_3 = 0 \quad \text{--- (ii)}$$

loop-3:

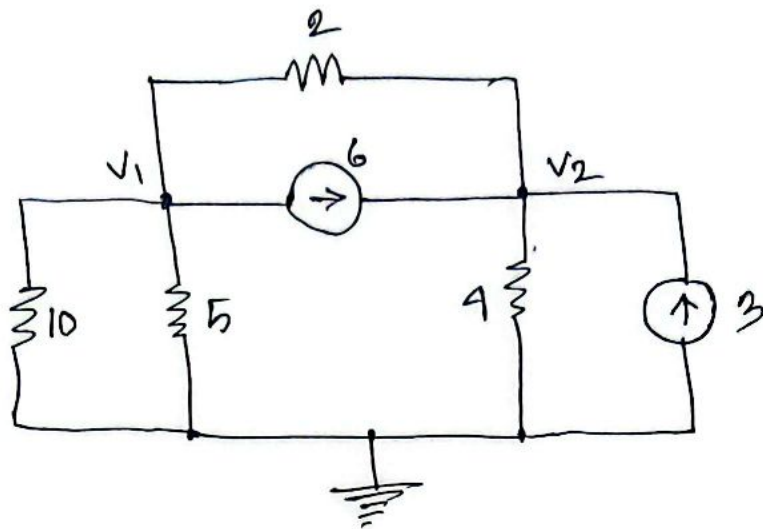
$$3.4I_1 + 0 + I_3 = 0 \quad \text{--- (iii)}$$

$$I_1 = -5, \quad I_2 = 16, \quad I_3 = 17$$

$$\therefore i = 16A,$$

$$V_o = I_1 \times 2 = -5 \times 2 = -10V$$

(ii)

Node - 1

$$\frac{V_1 - 0}{10} + \frac{V_1 - 0}{5} + \frac{V_1 - V_2}{2} + 6 = 0$$

$$\Rightarrow \frac{V_1 + 2V_1 + 5V_1 - 5V_2 + 60}{10} = 0$$

$$\Rightarrow 8V_1 - 5V_2 = -60 \quad \text{--- (i)}$$

Node - 2

$$\frac{V_2 - V_1}{2} + \frac{V_2}{4} = 6 + 3$$

$$\Rightarrow \frac{2V_2 - 2V_1 + V_2}{4} = 9$$

$$\Rightarrow -2V_1 + 3V_2 = 36 \quad \text{--- (ii)}$$

$$\therefore V_1 = 0V, \quad V_2 = 12V$$